# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

### *Solubility*

**1.** Which of the following halide is soluble in water?

(A) AgF (B) AgCl (C) AgBr (D) AgI

**CQA160**

### *Test of Carbonate and Bicarbonate*

**2.** A gas turns lime water milky and acidified $K_2Cr_2O_7$ solution green then gas is:

(A) HCl (B) $H_2S$ (C) $SO_2$ (D) $CO_2$

**CQA155**

### *Test of Sulphite ( $SO_3^{2\ominus}$ ); Bisulphite ( $HSO_3^{\ominus}$ )*

**3.** When a salt is heated with dilute $H_2SO_4$ and $KMnO_4$ solution, the pink colour of $KMnO_4$ is discharged, the salt is :

(A) a sulphite (B) a carbonate (C) a nitrate (D) a bicarbonate

**CQA145**

### *Test of Sulphide ($S^{2\ominus}$)*

**4.** $Pb(CH_3COO)_2$ gives.....colour with $H_2S$ :-

(A) Black (B) White (C) Red (D) Orange

**CQA190**

### *Test of Acetate ($CH_3COO^{\ominus}$)*

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**5.** When $CH_3COONa$ heated with solid $As_2O_3$ then compound X is formed. The smell of compound X is

(A) Pungent smell (B) Rotten Fish smell (C) Nausting smell (D) Rotten egg smell

**CQA094**

### *Test of Nitrite ( $NO_2^{\ominus}$ )*

**6.** A gas has smell like rotten egg and turns lead acetate paper black. The gas is :

(A) $NO_2$ (B) $H_2S$ (C) $CO_2$ (D) $SO_2$

**CQA156**

**7.** When KI is added to acidified solution of sodium nitrite :

(A) NO gas is liberated and $I_2$ is set free (B) $N_2$ gas is liberated and HI is produced

(C) $N_2O$ gas is liberated and $I_2$ is set free (D) $N_2$ gas is liberated and HOI is produced

**CQA159**

### *"Wet test of Acidic Radicals of Class A subgroup-II (Preliminary Tests)"*

**8.** Which of the following anion behaves in a different manner than other on heating with conc. $H_2SO_4$?

(A) $Cl^{\ominus}$ (B) $I^{\ominus}$

(C) $Br^{\ominus}$ (D) All behave in a similar manner

**CQA151**

### *Test of Chloride ($Cl^{\ominus}$), Bromide ($Br^{\ominus}$), Iodide ($I^{\ominus}$)*

**9.** When $K_2Cr_2O_7$ is heated with conc. $H_2SO_4$ and soluble chloride such as KCl :

(A) red vapours of $CrO_2Cl_2$ are evolved (B) $Cl^{\ominus}$ ion is oxidized to $Cl_2$ gas

(C) $CrCl_3$ is formed (D) $Cr_2O_7^{2\ominus}$ ion is reduced to green $Cr^{3\oplus}$ ion

**CQA148**



For More Join: @IITJEE_Advanced

**10.** Which of the following will not give positive chromyl chloride test ?

(A) Copper chloride, $CuCl_2$. (B) Mercuric chloride, $HgCl_2$.

(C) Zinc chloride, $ZnCl_2$. (D) Anilinium chloride $C_6H_5NH_3Cl$.

**CQA152**

### *Test of Nitrate ( $NO_3^{\ominus}$ )*

**11.** The brown ring test for $NO_2^{\ominus}$ and $NO_3^{\ominus}$ is due to formation of complex ion with formula :

(A) $[Fe(H_2O)_6]^{2\oplus}$ (B) $[Fe(CN)_5(NO)]^{2\ominus}$ (C) $[Fe(H_2O)_5(NO)]^{2\oplus}$ (D) $[Fe(H_2O)(NO)_5]^{2\oplus}$

**CQA149**

**12.** Comment True (T) or False (F) for the following statements.

(I) Same gas comes out when $Na_2S$ is treated with dil. HCl and $Na_2SO_3$ is treated with (Zn + dil. $H_2SO_4$)

(II) Same observation when acidified nitrite solution is treated with urea or thiourea followed by addition of $FeCl_3$ solution.

(III) Same observation when nitrite solution or acetate solution is treated with dil. $H_2SO_4$ solution

(IV) With $CaCl_2$ and $BaCl_2$ both acetate and formate do not react.

(A) FTTF (B) TFTT (C) TFFT (D) FTTT

**CQA099**

### *Test of Oxalate ( $C_2O_4^{2\ominus}$ )*

**13.** Which of the following pair of acidic radical can be distinguished by using dil $H_2SO_4$?

(I) $C_2O_4^{2\ominus}$ and $NO_3^{\ominus}$ (II) $NO_3^{\ominus}$ and $NO_2^{\ominus}$ (III) $Cl^{\ominus}$ and $Br^{\ominus}$ (IV) $HCO_3^{\ominus}$ and $CO_3^{2\ominus}$

(A) I and II (B) II only (C) II and IV (D) III and IV

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**CQA150**

**14.** Salt 'X' + dil. $H_2SO_4 \longrightarrow \underset{gas}{(B)} + H_2O$

Salt 'Y' + conc. $H_2SO_4 \longrightarrow \underset{gas}{(B)} + \underset{gas}{(C)} + H_2O$

Then salt 'X' and 'Y' may contain anions respectively :

(A) $SO_3^{2\ominus}, S^{2\ominus}$ (B) $C_2O_4^{2\ominus}, CO_3^{2\ominus}$ (C) $CO_3^{2\ominus}, C_2O_4^{2\ominus}$ (D) $SO_3^{2\ominus}, S_2O_3^{2\ominus}$

**CQA191**

### *Class (B) : Test of Phosphate ( $PO_4^{3\ominus}$ )*

**15.** A metal salt solution gives a yellow precipitate with silver nitrate. The precipitate dissolves in dil. Nitric acid as well as in ammonium hydroxide. The solution contains.

(A) $Br^{\ominus}$ (B) $I^{\ominus}$ (C) $PO_4^{3\ominus}$ (D) $SO_4^{2\ominus}$

**CQA140**

### *Classification of Basic Radicals*

**16.** Identify the **INCORRECT** group reagent for their respective cations .

(A) $Pb^{2\oplus}, Cu^{2\oplus}, As^{3\oplus}$, – dil. HCl + $H_2S$

(B) $Al^{3\oplus}, Fe^{3\oplus}, Cr^{3\oplus}$, – $NH_4Cl + NH_4OH$

(C) $Zn^{2\oplus}, Ni^{2\oplus}, Mn^{2\oplus}, Co^{2\oplus}$ – $NH_4Cl + NH_4OH + H_2S$

(D) $Ba^{2\oplus}, Cr^{2\oplus}, Ca^{2\oplus}$, – dil. HCl

**CQA192**



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### *Test of Basic Radicals: Group Zero*

**17.** Ammonium salt gives brown colour with alkaline solution of Nessler's reagent to form iodide of Millon's base. The formula of this compound is :-

(A) $NH_2–Hg–OHg–I$ (B) $NH_2–O–Hg–HgI$

(C) $K_2HgI_4$ (D) $NH_2–Hg–I$

**CQA193**

### *Test of Basic Radicals: Group 1*

**18.** A cold aqueous solution of $PbCl_2$ gives golden yellow precipitate on addition of

(A) KCl solution (B) KI solution (C) NaCl solution (D) $K_2SO_4$ solution

**CQA142**

### *Test of Basic Radicals: Group 2A*

**19.** Which of the following reagents turns white precipitate of AgCl yellow?

(A) $NaNO_3$ (B) $Na_3AsO_3$ (C) $Na_3AsO_4$ (D) NaCN

**CQA139**

**20.** When a metal sulphate is heated in dry test tube, the colour changes from blue to white. Then metal sulphate may be :

(A) $BaSO_4$ (B) $CuSO_4.5H_2O$ (C) $Na_2SO_4$ (D) None of these

**CQA153**

### *Test of Basic Radicals: Group 2B*

**21.** In which of the following pairs the precipitates are red and black coloured respectively and both precipitates are soluble in excess KI solution ?

(A) $HgI_2$, $Hg_2I_2$ (B) $HgI_2$, $BiI_3$ (C) $Cu_2I_2$, AgI (D) $CdI_2$, $PbI_2$

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**CQA123**

**22.** A 0.1 M solution of certain cation will precipitate with 0.1 M solutions of all these anions $OH^{\ominus}$, $CO_3^{2\ominus}$ & $SO_4^{2\ominus}$. This description is true for the cation.

(A) $Fe^{2\oplus}$ (B) $Mg^{2\oplus}$ (C) $Ba^{2\oplus}$ (D) $Pb^{2\oplus}$

**CQA158**

**23.** When bismuth chloride is poured into a large volume of water, the white precipitate produced is:

(A) $Bi(OH)_3$ (B) $Bi_2O_3$ (C) BiOCl (D) $Bi_2OCl_3$

**CQA194**

### *Test of Basic Radicals : Group 3*

**24.** Which of the following combines with Fe(II) ions to form a brown complex?

(A) $N_2O$ (B) NO (C) $N_2O_5$ (D) $N_2O_4$

**CQA136**

**25.** Sulphide ion reacts with $Na_2[Fe(CN)_5NO]$ to form a purple coloured compound (X). In this reaction oxidation state of iron.

(A) changes from +2 to +3 (B) changes from +3 to +2

(C) changes from +2 to +4 (D) does not change.

**CQA144**

**26.** $Fe(OH)_3$ and $Cr(OH)_3$ precipitates can be completely separated by :

(A) Aq. $NH_3$ (B) HCl (C) $NaOH/H_2O_2$ (D) $H_2SO_4$

**CQA119**



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**27.** An original solution of an inorganic salt in dilute HCl gives a brown colouration with potassium hexacyanidoferrate (III) and reddish-brown colouration with sodium acetate solution. The cation of the salt is :

(A) $Ni^{2\oplus}$ (B) $Fe^{3\oplus}$ (C) $Cu^{2\oplus}$ (D) none

**CQA127**

**28.** An inorganic lewis acid (X) fumes in moist air, and intensity of fumes increases when rod dipped in $NH_4OH$ is brought near to it. An acidic solution of (X) on addition of $NH_4Cl$ and $NH_4OH$ gives a precipitate which dissolves in NaOH solution. A solution of (X) does not give precipitate with $H_2S$ Hence, compound (X) is

(A) $FeCl_3$ (B) $AlCl_3$ (C) $CrCl_3$ (D) $ZnCl_2$

**CQA126**

**29.** Potassium thiocyanate solution reacts with ferric chloride to give :-

(A) Pink colour (B) Deep blue colour

(C) Green colour (D) Blood-red colour

**CQA195**

**30.** Fe is precipitated as $Fe(OH)_3$ and not $Fe(OH)_2$ because :-

(A) $Fe(OH)_2$ is not precipitated as it is soluble

(B) $Fe(OH)_3$ is soluble

(C) $Fe(OH)_2$ is insoluble

(D) None of these

**CQA196**

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**31.** $Fe_2(SO_4)_3$ solution + $Na_2S_2O_3$ solution.

Select correct about above reaction.

(A) Initially violet solution produces (B) Finally redox change is observed

(C) $S_2O_3^{2\ominus}$ is oxidised to $S_2O_6^{2\ominus}$ (D) All of the above

**CQA197**

**32.** $Fe^{3\oplus}$ ions produce deep blue colouration when treated with reagent (ft). R is :

(A) $K_3[Fe(CN)_6]$ (B) $K_4[Fe(CN)_6]$ (C) KSCN (D) excess of $NH_4OH$

**CQA198**

## *Test of Basic Radicals : Group 4*

**33.** $NiCl_2 + KCN \underset{\text{excess}}{\longrightarrow}$ Yellow colour solution $\xrightarrow{NaOH + Br_2 \text{ water}}$ "X" ppt

Colour of precipitate of " X " is :

(A) green (B) black (C) yellow (D) Reddish Brown

**CQA143**

**34.** $H_2S$ in the presence of HCl precipitates $II^{nd}$ group but not IV group because :

(A) HCl activates $H_2S$

(B) HCl increases concentration of $Cl^{\ominus}$

(C) HCl decreases concentration of $S^{2\ominus}$

(D) HCl lowers the solubility of $H_2S$ in solution

**CQA120**



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**35.** A coloured solution of a salt gives following reactions.
(i) It gives white precipitate with sodium hydroxide which becomes brown on exposure to air.
(ii) It gives light pink precipitate with $NH_4OH$ solution which is insoluble in excess $NH_3$.
(A) $Mn^{2\oplus}$ (B) $Zn^{2\oplus}$
(C) $Al^{3\oplus}$ (D) $Ni^{2\oplus}$

**CQA128**

**36.** Select compound which is written with its colour correctly.
(A) $K_2MnO_4$ (Purple) (B) CdS black)
(C) $Cr_2O_3$, (Green) (D) MnS (Black)

**CQA199**

**37.** Which pair of ions can not be distinguished by given reagent?

| **Pair of ions** | | **Reagent** |
|---|---|---|
| (A) $Fe^{2\oplus}$, $Fe^{3\oplus}$ | $\Rightarrow$ | $K_4[Fe(CN)_6]$(aq.) |
| (B) $CO_3^{2\ominus}$ , $HCO_3^{\ominus}$ | $\Rightarrow$ | $CaCl_2$ (aq.) |
| (C) $Hg^{2\oplus}$, $Ag^{\oplus}$ | $\Rightarrow$ | HCl (dil.) |
| (D) $Al^{3\oplus}$, $Zn^{2\oplus}$ | $\Rightarrow$ | NaOH (excess) |

**CQA200**

## *Test of Basic Radicals : Group 5*

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**38.** $Na_2SO_4 + BaCl_2 \longrightarrow$ White precipitate $\xrightarrow{\text{Fuse this precipitate on charcoal with } Na_2CO_3 \text{ and extract the soluble substance with } H_2O}$ Aqueous solution. $\xrightarrow{\text{Add dil } H_2SO_4 \text{ and heat the solution}}$ Gas 'G' is evolved.

The gas 'G' will show which of the following property ?
(A) It turns lead acetate filter paper black.
(B) It turns acidified $K_2Cr_2O_7$ filter paper green.
(C) It produces purple colouration on filter paper moistened with sodium nitroprusside already made alkaline with sodium hydroxide.
(D) All of these

**CQA157**

**39.** There are four test tubes containing dilute HCl, $BaCl_2$, $HgCl_2$ and $KNO_3$ solutions. Which of the following reagents will help in the identification of $BaCl_2$ ?
(A) NaOH (B) $K_2CrO_4$ (C) $AgNO_3$ (D) both (B) and (C)

**CQA147**

**40.** A salt on treatment with dil. HCl gives a pungent smelling gas and a yellow precipitate. The salt gives green flame when tested. The solution gives a yellow precipitate with potassium chromate. The salt is:
(A) $NiSO_4$ (B) $BaS_2O_3$ (C) $PbS_2O_3$ (D) $CuSO_4$

**CQA202**



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### *Dry tests for Basic Radicals: Flame test*

**41.** Metal **(M)** shows crimson red colour in flame test and its halide is deliquescent then metal **(M)** could be :

(A) Li (B) Mg (C) Ca (D) Ba

**CQA138**

**42.** A fire work gave bright crimson red light. It probably contained a salt of :

(A) Ca (B) Sr (C) Ba (D) Mg

**CQA141**

### *Borax Bead test, Microcosmic Bead Test*

**43.** Which one of the following ions does not give borax bead test ?

(A) $Cr^{3\oplus}$ (B) $Cu^{2\oplus}$ (C) $Mn^{2\oplus}$ (D) $Zn^{2\oplus}$

**CQA137**

**44.** In the borax bead test of $Co^{2\oplus}$, the blue colour of bead is due to the formation of:

(A) $B_2O_3$ (B) $Co_3B_2$ (C) $Co(BO_2)_2$ (D) CoO

**CQA111**

### *Heating Effect*

**45.** Which of the following can not evolve more than one gas (vapour) if heated in dry test tube.

(A) $NaNO_3(s)$ (B) $MgCO_3(s)$ (C) $FeSO_4(s)$ (D) $(NH_4)_2Cr_2O_7(s)$

**CQA154**

**46.** Which of the following leaves no residue on heating?

(A) $Pb(NO_3)_2$ (B) $NH_4NO_3$ (C) $Cu(NO_3)_2$ (D) $NaNO_3$

**CQA112**

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# EXERCISE - (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

### *Test of Sulphite ( $SO_3^{2\ominus}$ ); Bisulphite ( $HSO_3^{\ominus}$ )*

**1.** $H_2S$ and $SO_2$ can be distinguished by :

(A) Litmus paper (B) $MnO_4^{\ominus}/H^{\oplus}$ (C) $(CH_3COO)_2Pb$ (D) None of these

**CQA164**

### *Test of Basic Radicals : Group 1*

**2.** Ion(s) which do(es) not produce ppt. with excess of $CN^{\ominus}$ ions :

(A) $K^{\oplus}$ (B) $Pb^{2\oplus}$ (C) $Hg_2^{2\oplus}$ (D) $Ag^{\oplus}$

**CQA203**

### *Test of Basic Radicals: Group 2B*

**3.** Which of the following is/are correctly matched ?

(A) $BiI_3\downarrow \longrightarrow$ Black precipitate (B) $CuI\downarrow \longrightarrow$ White precipitate

(C) $PbI_2\downarrow \longrightarrow$ Yellow precipitate (D) $HgI_2\downarrow \longrightarrow$ Red precipitate

**CQA161**

**4.** Which of the following statement(s) is/are correct ?

(A) Yellow precipitated of silver arsenite is soluble in both nitric acid and ammonia.

(B) Potassium cyanide when added in very small quantity to copper sulphate solution, produces first yellow precipitate which quickly converts in to white precipitate.

(C) Black precipitate of $BiI_3$ turns orange on heating with water.

(D) White precipitate of $Bi(OH)_3$ turns yellowish brown, when boiled.

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**CQA163**

**5.** Which of the following statements is/are incorrect ?

(A) A filter paper moistened with cadmium acetate solution turns yellow, when brought in contact with $H_2S$ gas.

(B) Both carbonate ions as well as bicarbonate ions in the solutions, give reddish-brown precipitate with mercury(II) chloride.

(C) Sulphites in presence of zinc, reacts with dilute $H_2SO_4$ to liberate $SO_3$ gas.

(D) None of these

**CQA165**

**6.** Which anion can produce ppt. with lead acetate solution?

(A) $NO_3^{\ominus}$ (B) $Cl^{\ominus}$ (C) $CO_3^{2\ominus}$ (D) $S^{2\ominus}$

**CQA204**

### *Test of Basic Radicals : Group 4*

**7.** $Co^{2\oplus}$ + KCN (not in excess) $\longrightarrow$ precipitate.

Select the correct statement(s) with respect to the precipitate.

(A) It is yellow in colour.

(B) It is reddish-brown in colour.

(C) It dissolves in excess of the reagent forming a brown solution.

(D) It is obtained when brown solution (option (C)) is acidified with dilute HCl in the cold.

**CQA162**



For More Join: @IITJEE_Advanced

**8.** Which of the following combination will give a red colour or precipitate in an aqueous medium?

(A) $Fe^{3\oplus}$ + $SCN^{\ominus}$ (B) $Fe^{3\oplus}$ + $[Fe(SN)_6]^{3\ominus}$

(C) $Ni^{2\oplus}$ + dimethyl glyoxime (D) $Co^{2\oplus}$ + $SCN^{\ominus}$

**CQA134**

**9.** $CoCl_2 + KNO_2 + CH_3COOH \longrightarrow [X] + H_2O + KCl + CH_3COOK + NO$

(Unbalanced equation)

Which is / are correct for 'X'?

(A) X is a yellow crystalline solid which is insoluble in water.

(B) X is a green coloured compounds known as Rinman's green.

(C) IUPAC name of X is potassium hexanitrito –N– cobaltate (II)

(D) The compound X is an inner orbital complex.

**CQA205**

### *Test of Basic Radicals : Group 5*

**10.** KCN is used for separating :

(A) $Co^{2\oplus}$ and $Ni^{2\oplus}$ (B) $Mn^{2\oplus}$ and $Zn^{2\oplus}$ (C) $Ba^{2\oplus}$ and $Ca^{2\oplus}$ (D) $Cu^{2\oplus}$ and $Cd^{2\oplus}$

**CQA133**

### *Borax Bead test, Microcosmic Bead Test*

**11.** Which of the following can be detected by the borax bead test ?

(A) $Ni^{2\oplus}$ (B) $Co^{2\oplus}$ (C) $Pb^{\oplus}$ (D) $Hg_2^{2\oplus}$

**CQA121**

### *Heating Effect*

**12.** Which of the following metal carbonate will give of metal and oxygen on heating-

(A) $Ag_2CO_3$ (B) $HgCO_3$ (C) $(NH_4)_2CO_3$ (D) $PbCO_3$

**CQA097**

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**13.** Which of the following metal nitrate produce $NO_2$ on heating

(A) $Hg(NO_3)_2$ (B) $RbNO_3$ (C) $Pb(NO_3)_2$ (D) $Cu(NO_3)_2$

**CQA110**

**14.** Which of the following salts release reddish brown gas when heated in a dry test tube?

(A) $LiNO_3$ (B) $KNO_3$ (C) $Pb(NO_3)_2$ (D) $AgNO_3$

**CQA166**

**15.** Which of the following pairs produce same gas?

(A) ($Ca_3N_2$ + dil. HCl) and ($NH_4NO_2$ on heating)

(B) [$(NH_4)_2Cr_2O_7$ on heating] and ($NH_4NO_2$ on heating)

(C) ($NH_4NO_3$ on heating) and [$Hg(NO_3)_2$ on heating]

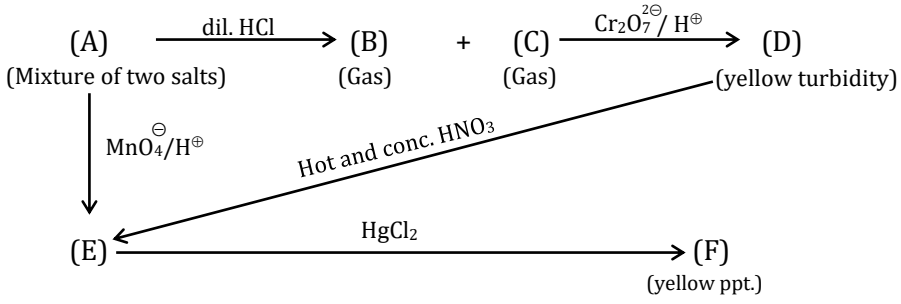
(D) ($NH_4Cl$ on heating) and [$NaNO_3$ + Zn + $NaOH_{(aq.)}$ on heating]

**CQA206**

## COMPREHENSION TYPE QUESTIONS

### *Class (B) : Test of Sulphate ( $SO_4^{2\ominus}$ )*

**Paragraph for Question No. 16 to 18**






For More Join: @IITJEE_Advanced

**16.** Find the anion(s) :

(A) $SO_3^{2\ominus}$ (B) $SO_3^{2\ominus}, S^{2\ominus}$ (C) $SO_3^{2\ominus}, SO_4^{2\ominus}$ (D) $S_2O_3^{2\ominus}$

**CQA173**

**17.** Find out (E) :

(A) $S^{2\ominus}$ (B) $CO_3^{2\ominus}$ (C) $S_2O_3^{2\ominus}$ (D) $SO_4^{2\ominus}$

**CQA174**

**18.** Find out (F) :

(A) $HgSO_4.2HgO$ (B) $HgSO_4.3HgO$ (C) $HgSO_4$ (D) $Hg_2SO_4.3HgO$

**CQA175**

## *Test of Basic Radicals : Group 4*

**Paragraph for Question No. 19 to 20**

Two students were given mixture of $Al^{3\oplus}$, $Cr^{3\oplus}$, $Fe^{3\oplus}$, $Zn^{2\oplus}$, $Ni^{2\oplus}$, $Co^{2\oplus}$ and $Mn^{2\oplus}$ ions as their soluble salt which were not interfering radicals. They performed the experiment as follows:

**Student I:** In the aqueous solution of ions he added first $NH_4Cl$ then $NH_4OH$ and then passed $H_2S$ gas. He got the precipitate (X).

**student II:** In the aqueous solution of ions he first added $NH_4Cl$ and then $NH_4OH$. He got the precipitate (Y) and a filtrate. In the filtrate, he passed $H_2S$ gas and again got a precipitate (Z).

Based on above information answer the following two questions:

**19.** Which of the following is not the precipitate (X) ?

(A) ZnS (B) MnS (C) $Cr(OH)_3$ (D) $Al_2S_3$

**CQA207**

**20.** Precipitate (Y) when treated with aqueous $Na_2O_2$ solution, gives a yellow colour solution due to formation of which of the following?

(A) ZnO (B) $Na_2CrO_4$ (C) $Na_2Cr_2O_4$ (D) $Fe(OH)_3$

**CQA208**

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## *Test of Basic Radicals : Group 5*

**Paragraph For Questions Q.21, Q.22 and Q.23 by appropriately matching the information given in the three columns of the following table.**

In Column-1 cation are given while in Column-2 reagent are given which is used for test and in Column-3 observation are given. [Consider each solution to be aqueous unless stated.]

| Column-1 | | Column-2 | | Column-3 | |
|---|---|---|---|---|---|
| (I) | $Bi^{3\oplus}$ | (i) | $NH_3$ | (P) | Black / white precipitate is obtained. |
| (II) | $Cu^{2\oplus}$ | (ii) | KI | (Q) | Blue / Brown precipitate is obtained. |
| (III) | $Fe^{3\oplus}$ | (iii) | KCN | (R) | Form soluble complex in excess reagent. |
| (IV) | $Zn^{2\oplus}$ | (iv) | $K_4[Fe(CN)_6]$ | (S) | Redox reaction take place. |

**21.** The only incorrect combination is:

(A) (I) (ii) (P) (B) (II) (ii) (S) (C) (II) (ii) (R) (D) (III) (iii) (R)

**CQA167**

**22.** The only correct combination is:

(A) (II) (ii) (R) (B) (III) (ii) (Q) (C) (III) (i) (R) (D) (II) (iii) (S)

**CQA168**

**23.** The only incorrect combination is:

(A) (III) (iv) (Q) (B) (IV) (ii) (P) (C) (I) (i) (P) (D) (IV) (i) (R)

**CQA169**



For More Join: @IITJEE_Advanced

## MATRIX MATCH TYPE QUESTION

### *Test of Sulphite ( $SO_3^{2\ominus}$ ); Bisulphite ( $HSO_3^{\ominus}$ )*

**24.** Match List-I with List-II and select the correct answer using the codes given below the lists :

| | **List-I** | | **List-II** |
|---|---|---|---|
| P. | White turbidity | 1. | $IO_3^{\ominus} + SO_2 + \text{starch} \longrightarrow$ |
| Q. | Rotten egg smell | 2. | $SO_2 + MnO_4^{\ominus} \longrightarrow$ |
| R. | Colourless solution | 3. | $Zn + NaOH + SO_2 \longrightarrow$ |
| S. | Blue colour | 4. | $CO_2 + Ca(OH)_2 \longrightarrow$ |

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 1 | 3 | 2 | 4 |
| (B) | 3 | 2 | 4 | 1 |
| (C) | 4 | 3 | 2 | 1 |
| (D) | 4 | 1 | 2 | 3 |

**CQA176**

### *Test of Basic Radicals : Group 5*

**25.** **Match list**

| List–I (Basic Radical) | | List–II (Group) | |
|---|---|---|---|
| P | $Al^{3\oplus}$ | 1 | II group |
| Q | $Zn^{2\oplus}$ | 2 | V group |
| R | $Ba^{2\oplus}$ | 3 | IV group |
| S | $Pb^{2\oplus}$ | 4 | III group |

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(A) P-4; Q-2; R-1; S-3 (B) P-2; Q-4; R-1; S-3 (C) P-3; Q-1; R-2; S-4 (D) P-4; Q-3; R-2; S-1

**CQA115**

### *Test of Basic Radicals : Group 6*

**26.** Match the column -

**Column-I**
(Radical which can decolorize acidic solution of $MnO_4^{\ominus}$ )

(A) Only acidic radical
(B) Only basic radical
(C) both radical
(D) neither acidic nor basic radical

**Column-II**
(Given salt)

(P) $Sn(NO_3)_2$
(Q) $KNO_2$
(R) $FeC_2O_4$
(S) $Na_2CO_3$
(T) $BaSO_3$

(A) (A) → (Q); (B) → (R); (C) → (R); (D) → (T)
(B) (A) → (Q,T); (B) → (P); (C) → (P); (D) → (S)
(C) (A) → (Q,T); (B) → (P); (C) → (R); (D) → (S)
(D) (A) → (Q); (B) → (P); (C) → (S); (D) → (T)

**CQA114**



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# EXERCISE - (S)

## Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

### *Test of Sulphide ($S^{2\ominus}$)*

**1.** $Na_2S + Na_2[Fe(CN)_5NO] \rightarrow$ "A" (Violet Color)
In Complex "A", number of type of ambidentate ligand is/are "a" and number of d-orbital involved in hybridisation is/are "b" Then 7a + 8 b will be :

**CQA185**

### *Test of Chloride ($Cl^{\ominus}$), Bromide ($Br^{\ominus}$), Iodide ($I^{\ominus}$)*

**2.** NaCl + Solid $K_2Cr_2O_7$ + Conc. $H_2SO_4 \longrightarrow$ "X" (reddish brown fumes)
How many axial-d-orbital are involved in hybridization of "X" ?

**CQA187**

### Test of Nitrate ( $NO_3^{\ominus}$ )

**3.** Find the total number of acidic radical which produce volatile product with dil HCl :
(i) $SO_4^{2\ominus}$ (ii) $I^{\ominus}$ (iii) $NO_2^{\ominus}$ (iv) $NO_3^{\ominus}$
(v) $SO_3^{2\ominus}$ (vi) $HCO_3^{\ominus}$

**CQA182**

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**4.** How many of the following liberate coloured vapour/gas with concentrated $H_2SO_4$?
KCl (s) + $K_2Cr_2O_7$ (s), $KNO_2$ (s), KI(s), KBr(s), KCl(s)
KBr (s) + $MnO_2$ (s), $KNO_3$, KCI(s) + $MnO_2$, $K_2SO_3$

**CQA186**

### *Test of Borate ( $BO_3^{3\ominus}$ )*

**5.** $BO_3^{3\ominus}$ + cone. $H_2SO_4 + CH_3 - CH_2 - OH \xrightarrow{\text{ignite}}$ (A)
Green flame
What is the oxidation number of central atom that is responsible for green flame in compound (A)?

**CQA184**

### *Test of Oxalate ( $C_2O_4^{2\ominus}$ )*

**6.** $Na_2CrO_4$ + dil.$H_2SO_4 \longrightarrow Na_2SO_4 + H_2CrO_4$
$H_2CrO_4 + 2H_2O_2 \xrightarrow{\text{amyl alcohol}}$ 'A' (blue colouration) + $H_2O$
then find a + b, if a = number of O – O bonds in "A"
b = number of Cr – O σ(sigma) bonds in "A"

**CQA183**

### *Test of Basic Radicals : Group 1*

**7.** An aqueous solution contains $Hg^{2\oplus}$, $Hg_2^{2\oplus}$, $Pb^{2\oplus}$, $Ag^{\oplus}$, $Bi^{3\oplus}$ and $Cd^{2\oplus}$. Out of these, how many ions will produce white precipitate with dilute HCl ?

**CQA210**



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### *Test of Basic Radicals: Group 2A*

**8.** $Cu^{2\oplus}$ + KCN (in excess) $\rightarrow$ soluble complex (X). How many statements are correct regarding complex (X) -
(i) the central atom has the co-ordination number of 6
(ii) the central atom has the co-ordination number of 4
(iii) the complex is sq.planar
(iv) the complex is diamagnetic
(v) the complex is paramagnetic

**CQA135**

### *Test of Basic Radicals: Group 2B*

**9.** Solution of $AsO_4^{3\ominus}$ ion containing considerable excess of $HNO_3$ on boiling with Ammonium molybdate test gives a yellow crystalline precipitate (X). Number of oxygen atom present in per molecule of precipitated (X) are :

**CQA179**

**10.** Oxidation state of Hg in Nessler's Reagent is :-

**CQA211**

### *Test of Basic Radicals : Group 3*

**11.** $Fe^{2\oplus}(aq) + NO_3^{\ominus}(aq) + H_2SO_4(conc) \longrightarrow$ Brown ring.
The oxidation number of iron in brown ring complex is :

**CQA181**

**12.** How many statements are correct ?
(i) Like $CO_3^{2\ominus}$, $SO_3^{2\ominus}$ also gives test with lime water.
(ii) In the test of $NO_3^{\ominus}$, brown ring is formed due to the formation of $[Fe(H_2O)_5 NO]^{2\oplus}$.
(iii) Aqueous solution of $Ag^{\oplus}$ ions gives brick red/red precipitate with $K_2CrO_4$.
(iv) Lead salts turn black on prolonged exposure in a chemical laboratory.
(v) In analysis of group II$^{nd}$ cations, $H_2S$ gas is passed in presence of hydrochloric acid to enhance the concentration of $S^{2\ominus}$ ions.
(vi) Aqueous solution of $Cu^{2\oplus}$ ions forms a green complex with $K_4[Fe(CN)_6]$.

**CQA188**

### *Test of Basic Radicals : Group 4*

**13.** An alcoholic solution of dimethylglyoxime is added to an aqueous solution of nickel(II) chloride. Slow addition of ammonium hydroxide led to the precipitation of a bright-red coloured metal complex.
Find out the number of hydrogen bonds present in the structure of the complex.

**CQA178**

**14.** Which of the following sulphides are not black in colour.
PbS, CdS, $As_2S_3$, $Sb_2S_3$, SnS, CoS, HgS, ZnS, MnS.

**CQA180**

### *Heating Effect*

**15.** On strong heating of $H_3PO_4$ and $H_3BO_3$, sum of oxidation number of P & B in the final product obtained is

**CQA122**

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**Part-2 : JEE Advanced (Decimal type)**

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

---

*Test of Thiosulphate ( $S_2O_3^{2\ominus}$ )*

**16.** In neutral or faintly alkaline solution, 4 moles permanganate anion quantitatively oxidise thiosulphate anions to produce X moles of a sulphur containing product. The magnitude of $\frac{X}{9}$ is:

**CQA212**

*Test of Chloride ($Cl^{\ominus}$), Bromide ($Br^{\ominus}$), Iodide ($I^{\ominus}$)*

**17.** $Fe^{2\oplus} + NO_3^{\ominus} + H_2SO_4(conc.) \rightarrow$ 'X' (Brown ring complex)

The magnetic moment of complex 'X' :

**CQA214**

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# EXERCISE - JEE (Main) PYQ

**1.** Arrange the following in increasing order of solubility product : **[JEE (Main) 2025]**
$Ca(OH)_2$, AgBr, PbS, HgS
(1) PbS < HgS < $Ca(OH)_2$ < AgBr (2) HgS < PbS < AgBr < $Ca(OH)_2$
(3) $Ca(OH)_2$ < AgBr < HgS < PbS (4) HgS < AgBr < PbS < $Ca(OH)_2$

**CQA223**

**2.** What are the products formed in sequence when excess of $CO_2$ is passed in slaked lime? **[JEE (Main) 2021]**
(1) $Ca(HCO_3)_2$, $CaCO_3$ (2) $CaCO_3$, $Ca(HCO_3)_2$
(3) CaO, $Ca(HCO_3)_2$ (4) CaO, $CaCO_3$

**CQA048**

**3.** Sodium extract is heated with concentrated $HNO_3$ before testing for halogens because : **[JEE (Main) 2016]**
(1) $S^{2\ominus}$and $CN^{\ominus}$, if present are decomposed by conc. $HNO_3$ and hence do not interfere in the test.
(2) Ag reacts faster with halides in acidic medium
(3) $Ag_2S$ and AgCN are soluble in acidic medium.
(4) Silver halides are totally insoluble in nitric acid.

**CQA041**

**4.** A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. nitric acid. The anion is : **[JEE (Main) 2018]**
(1) $S^{2\ominus}$ (2) $SO_4^{2\ominus}$ (3) $CO_3^{2\ominus}$ (4) $Cl^{\ominus}$

**CQA045**

**5.** Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown - red vapours of: **[JEE (Main) 2013]**
(1) $CrO_3$ (2) $Cr_2O_3$ (3) $CrCl_3$ (4) $CrO_2Cl_2$

**CQA031**

**6.** An organic compound 'A' is oxidized with $Na_2O_2$ followed by boiling with $HNO_3$.The resultant solution is then treated with ammonium molybdate to yield a yellow precipitate.
Based on above observation, the element present in the given compound is : **[JEE (Main) 2019]**
(1) Sulphur (2) Nitrogen (3) Fluorine (4) Phosphorus

**CQA047**

**7.** On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is **[JEE (Main) 2024]**
(1) $H_2S$ (2) $CO_2$ (3) $NH_3$ (4) $Cl_2$

**CQA215**

**8.** Consider the following equilibrium **[JEE (Main) 2013]**
$AgCl\downarrow + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^{\oplus} + Cl^{\ominus}$
White precipitate of AgCl appears on adding which of the following?
(1) $NH_3$ (2) Aqueous NaCl
(3) Aqueous $NH_4Cl$ (4) Aqueous $HNO_3$

**CQA035**

**9.** Identify A, B and C in the given below reaction sequence **[JEE (Main) 2025]**

$$\boxed{A} \xrightarrow{HNO_3} Pb(NO_3)_2 \xrightarrow{H_2SO_4} \boxed{B} \xrightarrow[(3)\ K_2CrO_4]{(1)\ \text{Ammonium acetate}\ (2)\ \text{Acetic acid}} \underset{\text{Yellow ppt}}{\boxed{C}}$$

(1) $PbCl_2$, $PbSO_4$, $PbCrO_4$ (2) PbS, $PbSO_4$, $PbCrO_4$
(3) PbS, $PbSO_4$, $Pb(CH_3COO)_2$ (4) $PbCl_2$, $Pb(SO_4)_2$, $PbCrO_4$

**CQA224**



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**10.** Identify incorrect statement **[JEE (Main) 2013]**

(1) Copper (I) compounds are colourless except where colour results from charge transfer

(2) Copper (I) compounds are diamagnetic

(3) $Cu_2S$ is black

(4) $Cu_2O$ is colourless

**CQA034**

**11.** Match **List I** with **List II**

| List-I (Compound) | | List-II (Colour) | |
|---|---|---|---|
| A | $Fe_4[Fe(CN)_6]_3.xH_2O$ | I. | Violet |
| B. | $[Fe(CN)_5NOS]^{4\ominus}$ | II. | Blood Red |
| C. | $[Fe(SCN)]^{2\oplus}$ | III. | Prussian Blue |
| D. | $(NH_4)_3PO_4.12MoO_3$ | IV. | Yellow |

Choose the **correct** answer from the options given below : **[JEE (Main) 2024]**

(1) A-III, B-I, C-II, D-IV (2) A-IV, B-I, C-II, D-III

(3) A-II, B-III, C-IV, D-I (4) A-I, B-II, C-III, D-IV

**CQA217**

**12.** An aqueous solution of a salt X turns blood red on treatment with $SCN^{\ominus}$ and blue on treatment with $K_4[Fe(CN)_6]$. X also gives a positive chromyl chloride test. The salt X is : **[JEE (Main) 2015]**

(1) $CuCl_2$ (2) $Cu(NO_3)_2$ (3) $FeCl_3$ (4) $Fe(NO_3)_3$

**CQA037**

**13.** Find the compound 'A' from the following reaction sequences. **[JEE (Main) 2025]**

$$A \xrightarrow{\text{aqua-regia}} B \xrightarrow[(2)AcOH]{(1)KNO_2|NH_4OH} \text{yellow ppt}$$

(1) ZnS (2) CoS (3) MnS (4) Nis

**CQA225**

**14.** A reagent which gives brilliant red precipitate with Nickel ions in basic medium is **[JEE (Main) 2024]**

(1) sodium nitroprusside (2) neutral $FeCl_3$

(3) meta-dinitrobenzene (4) dimethyl glyoxime

**CQA218**

**15.** During the qualitative analysis of salt with cation $y^{2\oplus}$, addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation ($y^{2\oplus}$) present respectively are: **[JEE (Main) 2022]**

(1) Dimethylglyoxime and $Ni^{2\oplus}$ (2) Dimethylglyoxime and $CO^{2\oplus}$

(3) Nessler's reagent and $Hg^{2\oplus}$ (4) Nessler's reagent and $Ni^{2\oplus}$

**CQA049**

**16.** The **incorrect** statement is : **[JEE (Main) 2018]**

(1) $Cu^{2\oplus}$ ion gives chocolate coloured precipitate with potassium ferrocyanide solution.

(2) $Cu^{2\oplus}$ and $Ni^{2\oplus}$ ions give black precipitate with $H_2S$ in presence of HCl solution.

(3) Ferric ion gives blood red colour with potassium thiocyanate.

(4) $Cu^{2\oplus}$ salts give red coloured borax bead test in reducing flame.

**CQA044**

**17.** The cation that will not be precipitated by $H_2S$ in the presence of dil. HCl is: **[JEE (Main) 2015]**

(1) $Pb^{2\oplus}$ (2) $As^{3\oplus}$ (3) $Co^{2\oplus}$ (4) $Cu^{2\oplus}$

**CQA036**

**18.** When concentrated HCl is added to an aqueous solution of $CoCl_2$, its colour changes from reddish pink to deep blue. Which complex ion gives blue colour in this reaction? **[JEE (Main) 2015]**

(1) $[CoCl_4]^{2\ominus}$ (2) $[CoCl_6]^{3\ominus}$ (3) $[CoCl_6]^{4\ominus}$ (4) $[Co(H_2O)_6]^{2\oplus}$

**CQA038**

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**19.** Which of the following statements is incorrect? **[JEE (Main) 2013]**

(1) $Fe^{2\oplus}$ ion also gives blood red colour with $SCN^{\ominus}$ ion

(2) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[Cu(NH_3)_4]^{2\oplus}$ ion.

(3) $Fe^{3\oplus}$ ion gives blood red colour with $SCN^{\ominus}$ ion.

(4) On passing $H_2S$ into $Na_2ZnO_2$ solution, a white ppt of ZnS is formed.

**CQA033**

**20.** Sodium Carbonate cannot be used in place of $(NH_4)_2CO_3$ for the identification of $Ca^{2\oplus}$, $Ba^{2\oplus}$ and $Sr^{2\oplus}$ ions (in group V) during mixture analysis because : **[JEE (Main) 2013]**

(1) Sodium ions will react with acid radicals

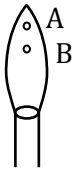
(2) Concentration of $CO_3^{2\ominus}$ ions is very low

(3) $Mg^{2\oplus}$ ions will also be precipitated

(4) $Na^{\oplus}$ ions will interfere with the detection of $Ca^{2\oplus}$, $Ba^{2\oplus}$ and $Sr^{2\oplus}$ ions

**CQA032**

**21.**

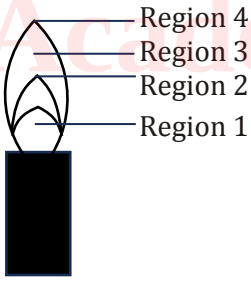



In a borax bead test under hot condition, a metal salt (one from the given) is heated at point B of the flame, resulted in green colour salt bead. The
spin-only magnetic moment value of the salt is ..... BM (Nearest integer)
[Given atomic number of Cu = 29, Ni = 28, Mn = 25, Fe = 26] **[JEE (Main) 2024]**

**CQA219**

**22.** The hottest region of Bunsen flame shown in the figure below is : **[JEE (Main) 2016]**

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(1) region 4 (2) region 1 (3) region 2 (4) region 3

**CQA040**

**23.** The metal ion whose electronic configuration is not affected by the nature of the ligand and which gives a violet colour in non-luminous flame under hot condition in borax bead test is **[JEE (Main) 2025]**

(1) $Ti^{3+}$ (2) $Ni^{2+}$ (3) $Mn^{2+}$ (4) $Cr^{3+}$

**CQA226**

**24.** A pink coloured salt turns blue on heating. The presence of which cation is most likely ? **[JEE (Main) 2015]**

(1) $Co^{2\oplus}$ (2) $Cu^{2\oplus}$ (3) $Zn^{2\oplus}$ (4) $Fe^{2\oplus}$

**CQA039**



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# EXERCISE - JEE (Advanced) PYQ

**1.** In the following reaction sequence in aqueous solution, the species X, Y and Z, respectively, are

$$S_2O_3^{2\ominus} \xrightarrow{Ag^{\oplus}} \underset{\text{Clear solution}}{X} \xrightarrow{Ag^{\oplus}} \underset{\text{white precipitate}}{Y} \xrightarrow{\text{with time}} \underset{\text{black precipitate}}{Z}$$

[JEE (Advanced) 2016]

(A) $[Ag(S_2O_3)_2]^{3\ominus}$, $Ag_2S_2O_3$, $Ag_2S$ (B) $[Ag(S_2O_3)_3]^{5\ominus}$, $Ag_2SO_3$,$Ag_2S$
(C) $[Ag(SO_3)_2]^{3\ominus}$, $Ag_2S_2O_3$, Ag (D) $[Ag(SO_3)_3]^{3\ominus}$, $Ag_2SO_4$,Ags

CQA076

**2.** A mixture of two salts is used to prepare a solution **S**, which gives the following results :

$$\underset{\text{precipitate(s) only}}{\text{White}} \xleftarrow[\text{Room temperature}]{\text{Dilute NaOH(aq.)}} \underset{\text{of the salts)}}{\mathbf{S}\ (\text{aq.solution}} \xrightarrow[\text{Room temperature}]{\text{Dilute HCl(aq.)}} \underset{\text{only}}{\text{White precipitate(s)}}$$

The correct option(s) for the salt mixture is(are) [JEE (Advanced) 2021]

(A) $Pb(NO_3)_2$ and $Zn(NO_3)_2$ (B) $Pb(NO_3)_2$ and $Bi(NO_3)_3$
(C) $AgNO_3$ and $Bi(NO_3)_3$ (D) $Pb(NO_3)_2$ and $Hg(NO_3)_2$

CQA082

**3.** With reference to aqua regia, choose the correct option(s). [JEE (Advanced) 2019]

(A) Reaction of gold with aqua regia produces $NO_2$ in the absence of air
(B) Aqua regia is prepared by mixing conc. HCl and conc. $HNO_3$ in 3 : 1 (v/v) ratio
(C) Reaction of gold with aqua regia produces an anion having Au in +3 oxidation state
(D) The yellow colour of aqua regia is due to the presence of NOCl and $Cl_2$

CQA079

**4.** A colorless aqueous solution contains nitrates of two metals, **X** and **Y**. When it was added to an aqueous solution of NaCl, a white precipitate was formed. This precipitate was found to be partly soluble in hot water to give a residue **P** and a solution **Q**. The residue **P** was soluble in aq. $NH_3$ and also in excess sodium thiosulfate. The hot solution **Q** gave a yellow precipitate with KI. The metals **X** and **Y**, respectively, are [JEE (Advanced) 2020]

(A) Ag and Pb (B) Ag and Cd (C) Cd and Pb (D) Cd and Zn

CQA080

**Paragraph for Question No. Q. 5 & 6**

An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl, gave a precipitate (P) and a filtrate (Q). The precipitate P was found to dissolve in hot water. The filtrate (Q) remained unchanged, when treated with $H_2S$ in a dilute mineral acid medium. However, it gave a precipitate (R) with $H_2S$ in an ammoniacal medium. The precipitate R gave a coloured solution (S), when treated with $H_2O_2$ in an aqueous NaOH medium.

**5.** The precipitate P contains [JEE (Advanced) 2013]

(A) $Pb^{2\oplus}$ (B) $Hg_2^{2\oplus}$ (C) $Ag^{\oplus}$ (D) $Hg^{2\oplus}$

CQA072

**6.** The coloured solution S contains [JEE (Advanced) 2013]

(A) $Fe_2(SO_4)_3$ (B) $CuSO_4$ (C) $ZnSO_4$ (D) $Na_2CrO_4$

CQA073

**Paragraph for Question No. Q.7 & 8**

The reaction of $K_3[Fe(CN)_6]$ with freshly prepared $FeSO_4$ solution produces a dark blue precipitate called Turnbull's blue. Reaction of $K_4[Fe(CN)_6]$ with the $FeSO_4$ solution in complete absence of air produces a white precipitate **X**, which turns blue in air. Mixing the $FeSO_4$ solution with $NaNO_3$, followed by a slow addition of concentrated $H_2SO_4$ through the side of the test tube produces a brown ring.



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**7.** Precipitate X is [JEE (Advanced) 2021]

(A) $Fe_4[Fe(CN)_6]_3$ (B) $Fe[Fe(CN)_6]$ (C) $K_2Fe[Fe(CN)_6]$ (D) $KFe[Fe(CN)_6]$

CQA083

**8.** Among the following, the brown ring is due to the formation of [JEE (Advanced) 2021]

(A) $[Fe(NO)_2(SO_4)_2]^{2\ominus}$ (B) $[Fe(NO)_2(H_2O)_4]^{3\oplus}$

(C) $[Fe(NO)_4(SO_4)_2]$ (D) $[Fe(NO)(H_2O)_5]^{3\oplus}$

CQA084

**Paragraph for Question No. 9 and 10**

When potassium iodide is added to an aqueous solution of potassium ferricyanide, a reversible reaction is observed in which a complex **P** is formed. In a strong acidic medium, the equilibrium shifts completely towards **P**. Addition of zinc chloride to **P** in a slightly acidic medium results in a sparingly soluble complex **Q**.

**9.** The number of moles of potassium iodide required to produce two moles of **P** is _____. [JEE (Advanced) 2024]

CQA220

**10.** The number of zinc ions present in the molecular formula of **Q** is _____. [JEE (Advanced) 2024]

CQA221

**11.** In the scheme given below, **X** and **Y**, respectively, are [JEE (Advanced) 2023]

Metal halide $\xrightarrow{\text{aq. NaOH}}$ White precipitate **(P)** + Filtrate **(Q)**

**P** $\xrightarrow[\text{heat}]{\text{aq.}H_2SO_4,\ PbO_2\text{(excess)}}$ **X** (a coloured species in solution)

**Q** $\xrightarrow[\text{warm}]{MnO(OH)_2,\ \text{Conc.}H_2SO_4}$ **Y** (gives blue-coloration with KI-starch paper)

(A) $CrO_4^{2\ominus}$ and $Br_2$ (B) $MnO_4^{2\ominus}$ and $Cl_2$ (C) $MnO_4^{\ominus}$ and $Cl_2$ (D) $MnSO_4$ and HOCl

CQA222

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**12.** The correct option(s) to distinguish nitrate salts of $Mn^{2\oplus}$ and $Cu^{2\oplus}$ taken separately is (are) [JEE (Advanced) 2018]

(A) $Mn^{2\oplus}$ show the characteristic green colour in the flame test

(B) Only $Cu^{2\oplus}$ show the formation of precipitate by passing $H_2S$ in acidic medium

(C) Only $Mn^{2\oplus}$ show the formation of precipitate by passing $H_2S$ in faintly basic medium

(D) $Cu^{2\oplus}/Cu$ has higher reduction potential than $Mn^{2\oplus}/Mn$ (measured under similar conditions)

CQA077

**13.** Among PbS, CuS, HgS, MnS, $Ag_2S$, NiS, CoS, $Bi_2S_3$ and $SnS_2$, the total number of BLACK coloured sulphides is : [JEE (Advanced) 2014]

CQA074

**14.** Upon treatment with ammoniacal $H_2S$, the metal ion that precipitates as a sulphide is : [JEE (Advanced) 2013]

(A) Fe(III) (B) Al (III) (C) Mg(II) (D) Zn(II) CQA071

**15.** The pair(s) of ions where BOTH the ions are precipitated upon passing $H_2S$ gas in presence of dilute HCl, is(are) [JEE (Advanced) 2015]

(A) $Ba^{2\oplus}$, $Zn^{2\oplus}$ (B) $Bi^{3\oplus}$, $Fe^{3\oplus}$ (C) $Cu^{2\oplus}$, $Pb^{2\oplus}$ (D) $Hg^{2\oplus}$, $Bi^{3\oplus}$

CQA075

**16.** The green colour produced in the borax bead test of a chromium(III) salt is due to – [JEE (Advanced) 2019]

(A) $Cr(BO_2)_3$ (B) CrB (C) $Cr_2(B_4O_7)_3$ (D) $Cr_2O_3$

CQA078

**17.** Thermal decomposition of $AgNO_3$ produces two paramagnetic gases. The total number of electrons present in the antibonding molecular orbitals of the gas that has the higher number of unpaired electrons is _____. [JEE (Advanced) 2022]

CQA085



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# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A | C | A | A | C | B | A | A | A | B |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | C | C | B | C | C | D | A | B | B | B |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | B | D | C | B | D | C | B | B | D | A |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | D | B | B | C | A | C | D | D | B | B |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | | | | |
| Ans. | A | B | D | C | B | B | | | | |

## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | ABC | AD | ABCD | ABC | BC | BCD | BCD | AC | AD | AD |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | AB | AB | ACD | ACD | BD | B | D | A | D | B |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | | | | |
| Ans. | C | D | B | C | D | C | | | | |

Rankers Academy JEE

## EXERCISE (S)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** | 30 | **2.** | 0 | **3.** | 3 | **4.** | 7 | **5.** | 3 |
| **6.** | 7 | **7.** | 3 | **8.** | 2 | **9.** | 40 | **10.** | 2 |
| **11.** | 1 | **12.** | 4 | **13.** | 2 | **14.** | 6 | **15.** | 8 |
| **16.** | 0.33 | **17.** | 3.874 | | | | | | |

## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 2 | 2 | 1 | 4 | 4 | 4 | 3 | 4 | 2 | 4 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 1 | 3 | 2 | 4 | 1 | 2 | 3 | 1 | 1 | 3 |
| Que. | 21 | 22 | 23 | 24 | | | | | | |
| Ans. | 6 | 3 | 2 | 1 | | | | | | |

## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A | AB | BCD | A | A | D | C | D | 2 | 3 or 2 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | | | |
| Ans. | C | BD | 7 | D | CD | A | 6 | | | |



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